Chapter 11: The Normal Distributions

The Practice of Statistics in the Life Sciences
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Part II: Probability and Distributions

Why Study Normal Distributions

Normal distributions are among the most important continuous probability models in introductory statistics.

Many measurements in the life sciences are reasonably described by a symmetric distribution with most observations near the centre and fewer observations farther away. Examples include some heights, some laboratory measurements and some measurement errors.

Chapter Goal

This chapter connects four ideas: the shape of a Normal curve, the area under the curve, z-scores and probability calculations.

What Is a Normal Curve

A Normal curve is a density curve that is

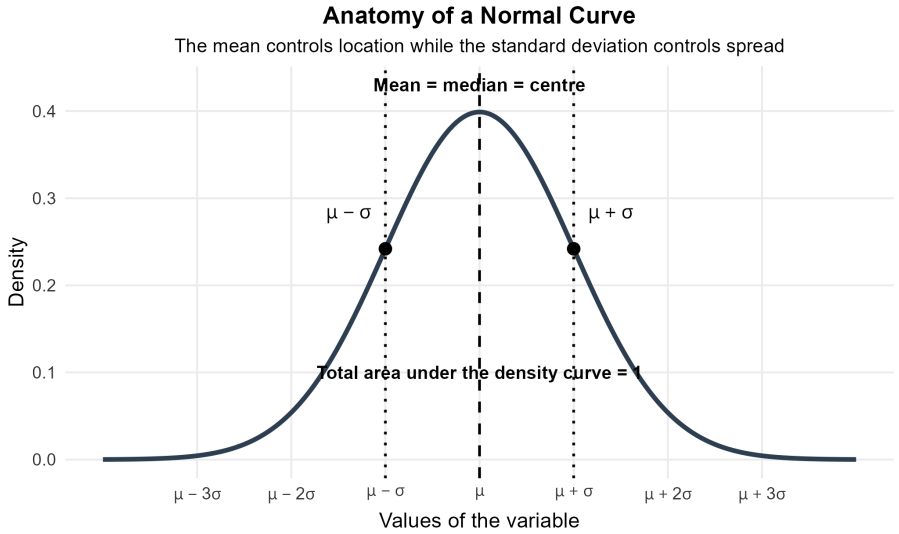
- symmetric
- single-peaked
- bell-shaped

Every Normal distribution is determined by two population parameters.

- The mean μ determines the location of the centre
- The standard deviation σ determines the spread

The total area under the curve is 1 so area can be (and is) interpreted as probability.

Anatomy of a Normal Curve



Key Geometric Features

Several geometric features are worth remembering.

1. Symmetry around μ means the mean and median are equal
2. The two tails approach the horizontal axis without touching it
3. Equal distances to the left and right of the mean enclose equal areas
4. The points $\mu - \sigma$ and $\mu + \sigma$ are the inflection points

Once μ and σ are known, the shape of the Normal curve is completely determined.

Location and Spread

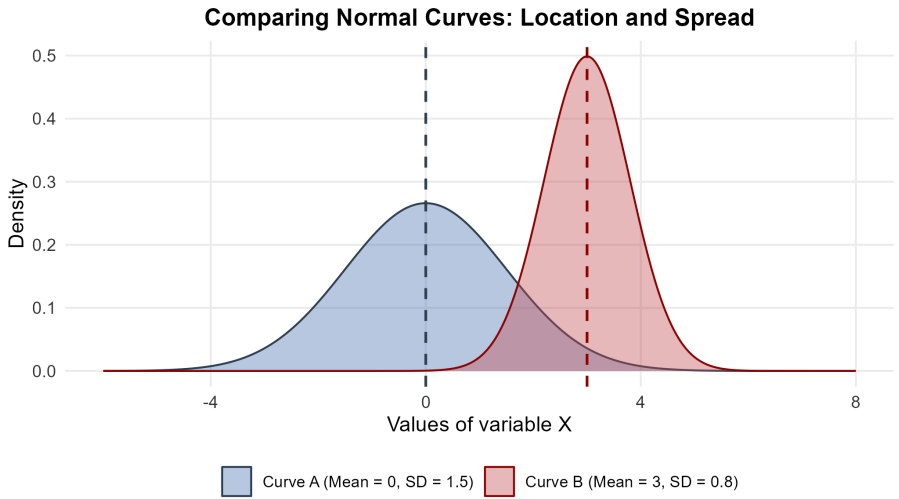
The mean moves the curve left or right without changing its bell shape.

The standard deviation changes the width of the curve.

- A larger standard deviation produces a wider and flatter curve
- A smaller standard deviation produces a narrower and taller curve

The taller appearance does not mean more probability.
Every density curve still has total area 1.

Location and Spread Visualized



Why Is the Narrower Curve Taller

A density curve must always have total area 1.

When the distribution is spread across a wider range of values, the height must decrease to keep the total area fixed.

When the distribution is concentrated over a narrower range of values, the peak must rise.

Key Idea

Width and height work together so that the total area under the curve stays equal to 1.

Inflection Points and the Standard Deviation

The standard deviation also has a geometric interpretation.

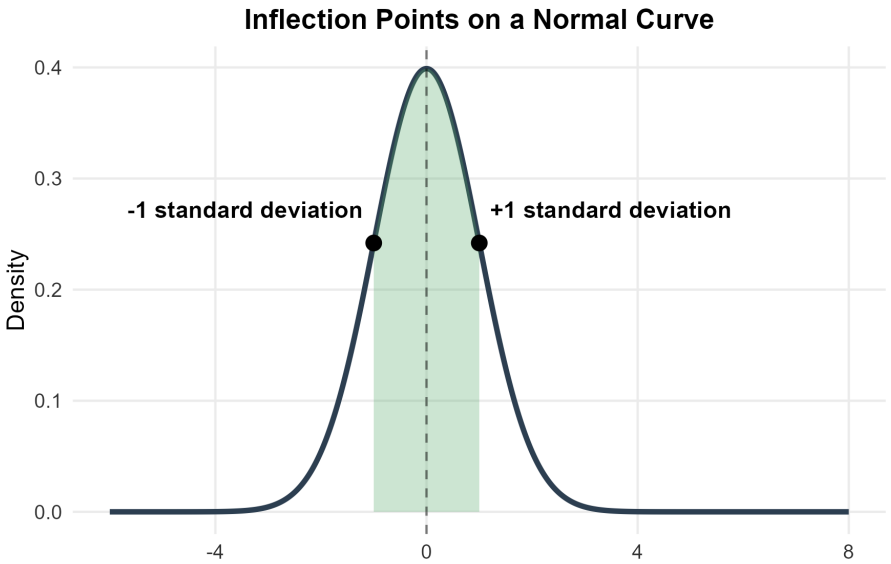
An **inflection point** is a point where the direction of curvature changes. In every Normal curve, the inflection points occur exactly one standard deviation from the mean.

$$\mu - \sigma$$

$$\mu + \sigma$$

This property is special to the Normal family and should not be applied automatically to other distributions.

Inflection Points on the Curve



Applications of Normal Models

- **Modelling continuous measurements**: Researchers can estimate the proportion of newborns whose birth weights fall within a specified range.
- **Modelling random measurement errors**: In regression, residuals help assess a Normal model for the errors, supporting confidence intervals and hypothesis tests.

These applications require checking whether a Normal model is reasonable for the relevant population or errors.

- **Approximating sampling distributions**: Ecologists can test whether a population's mean body mass differs from a reference value.
- **Comparing observations through standardization**: Laboratories can express measurements from different instruments in standard deviation units.

Standardization gives a common scale. A Normal model also allows us to translate standardized values into percentiles.

The 68–95–99.7 Rule

Every Normal distribution has the same approximate area pattern when distances are measured in standard-deviation units.

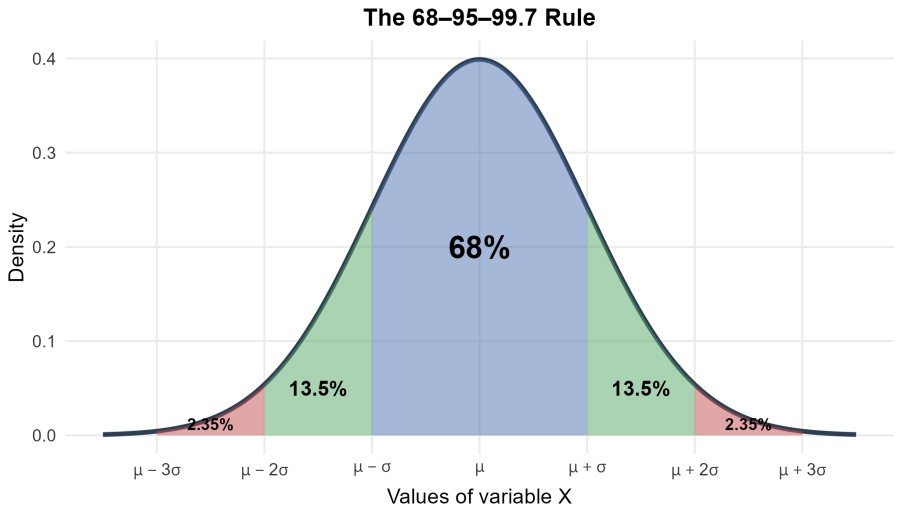
$$P(\mu - \sigma < X < \mu + \sigma) \approx 0.68$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997$$

These are approximations and not exact probabilities.

The 68–95–99.7 Rule



Using Symmetry to Find Tail Areas

The rule gives several other useful areas.

- The middle 68% leaves 32% outside, so each tail beyond $\pm 1\sigma$ contains about 16%
- The middle 95% leaves 5% outside, so each tail beyond $\pm 2\sigma$ contains about 2.5%
- The middle 99.7% leaves 0.3% outside, so each tail beyond $\pm 3\sigma$ contains about 0.15%

This rule is most useful when a boundary is exactly one, two or three standard deviations from the mean.

Example 11.1: Heights of Patients

Question

Suppose the heights of adult female patients aged 18 to 24 in a clinic are approximately Normally distributed with mean 64.5 inches and standard deviation 2.5 inches. What interval contains approximately the middle 95% of heights, and what proportion lies above the upper end of that interval?

Explanation

Approximately 95% of a Normal distribution lies within two standard deviations of the mean. The remaining 5% is split equally between the two tails because the curve is symmetric.

Example 11.1: Heights of Patients

Starting with

$$P(\mu - 2\sigma < X < \mu + 2\sigma)$$

The lower boundary is

$$\mu - 2\sigma = 64.5 - 2(2.5) = 59.5 \text{ inches}$$

The upper boundary is

$$\mu + 2\sigma = 64.5 + 2(2.5) = 69.5 \text{ inches}$$

Approximately 95% of the patients' heights lie between 59.5 inches and 69.5 inches.
The upper tail contains half of the remaining 5%, so

$$P(\text{height} > 69.5) = \frac{0.05}{2} = \boxed{0.025}$$

Under this model, about 2.5% of the patients are taller than 69.5 inches.

Example 11.2: Heights of Patients

Let H be the height, in inches, of a randomly selected adult female patient from the clinic. The Normal model has mean $\mu = 64.5$ inches and standard deviation $\sigma = 2.5$ inches.

Question

What is the approximate probability that the patient is taller than 62 inches? That is, find

$$P(H > 62)$$

Explanation

Since $62 = 64.5 - 2.5 = \mu - \sigma$, we need the area to the right of one standard deviation below the mean.

Example 11.2: Heights of Patients

The middle 68% lies between $\mu - \sigma$ and $\mu + \sigma$

$$P(62 < H < 67) = 0.68 \quad \text{approximately}$$

The area above $\mu + \sigma = 67$ is approximately

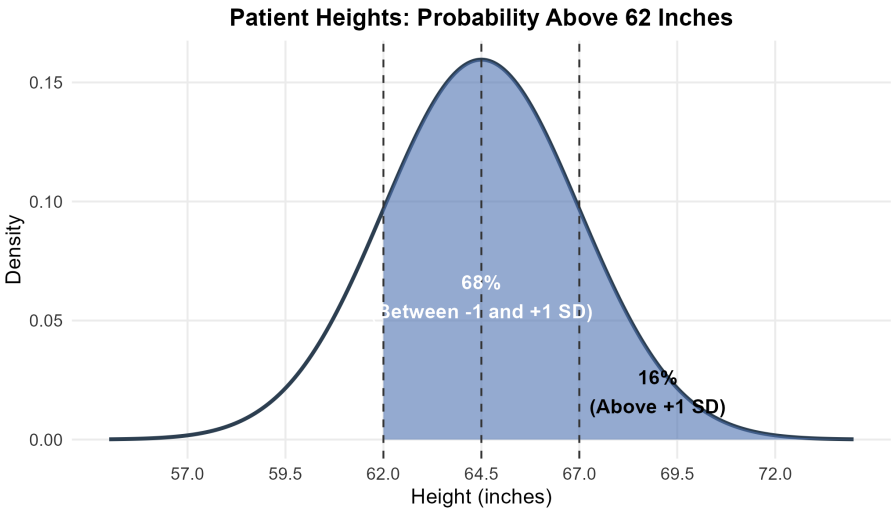
$$P(H > 67) = \frac{1 - 0.68}{2} = 0.16$$

Adding these disjoint regions gives

$$P(H > 62) = P(62 < H < 67) + P(H > 67) \approx 0.68 + 0.16 = \boxed{0.84}$$

Under this model, approximately 84% of the female patients are taller than 62 inches.

Example 11.2: Heights of Patients



Notation for a Normal Distribution

We can write the a normal random variable in compact notation as $X \sim N(\mu, \sigma)$
In these notes the second parameter is the **standard deviation**.

For the patient height model

$$H \sim N(64.5, 2.5)$$

This means that H follows a Normal distribution with mean 64.5 inches and standard deviation 2.5 inches.

A Normal Distribution Is an Approximation

Real data are never mathematically perfect.

A Normal model is still useful when it gives a good approximation to the shape of the distribution. The decision to use a Normal model should come from the data and the context rather than from habit.

Approximate models are common in statistics. Their value comes from being useful, interpretable and reasonably accurate for the purpose at hand.

Standardizing and Z-Scores

A **z-score** measures how far an observation lies from the mean in standard-deviation units.

$$z = \frac{x - \mu}{\sigma}$$

- A positive z-score means the observation is above the mean.
- A negative z-score means the observation is below the mean.
- A z-score of 0 means the observation is exactly at the mean.

Example 11.3: Standardizing Heights of Patients

Question

Patient heights are modelled as approximately Normal with mean 64.5 inches and standard deviation 2.5 inches. What are the z-scores for 70 inches and 60 inches?

Explanation

The standardizing formula is used for each height. The resulting z-score tells us how many standard deviations that height lies above or below the mean.

Example 11.3: Standardizing Heights of Patients

For 70 inches

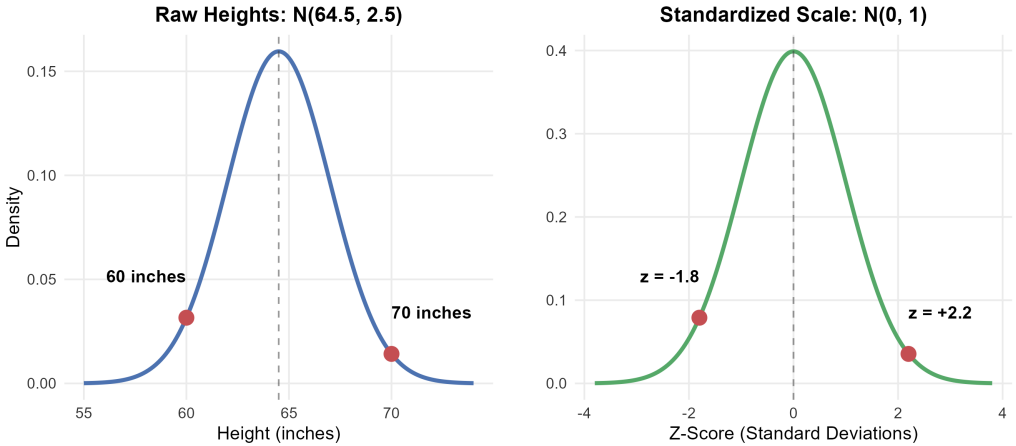
$$z = \frac{70 - 64.5}{2.5} = 2.2$$

For 60 inches

$$z = \frac{60 - 64.5}{2.5} = -1.8$$

A height of 70 inches is 2.2 standard deviations above the mean. A height of 60 inches is 1.8 standard deviations below the mean.

Standardizing Women's Heights Visualized



Why Standardize

Standardizing places observations from different settings on a common scale.

A raw value such as 28 points on one test cannot be directly compared with 1250 points on another test. Their z-scores can be compared because both are measured in standard-deviation units from their own means.

Standardization changes centre and scale. It does not magically make non-Normal data become Normal.

The Standard Normal Distribution

When a Normal random variable is standardized, the result follows the **standard Normal distribution**.

$$Z \sim N(0, 1)$$

This distribution has mean 0 and standard deviation 1.
Once values are converted to z-scores, every Normal-distribution problem can be related to the same common curve.

Applications of Z-Scores and the 68–95–99.7 Rule

The ideas in this section are not only theoretical.

Z-scores and the empirical rule are used to

- assess whether a measurement is unusually low or unusually high
- compare observations from different scales
- set review thresholds and safety limits
- interpret standard scores such as T-scores

Application: Assessing Patient Health

Question

Healthy adult male hemoglobin levels are modelled as Normal with mean 15.0 g/dL and standard deviation 1.0 g/dL. A patient has hemoglobin 13.0 g/dL. What is the z-score and what proportion of healthy men have a lower level?

Solution

$$z = \frac{13.0 - 15.0}{1.0} = -2.0$$

A z-score of -2.0 is two standard deviations below the mean. By the 68–95–99.7 rule, about 95% lies within $\pm 2\sigma$, leaving about 5% outside. The lower tail therefore contains about 2.5%.

Application: Quality Control of Solutions

Question

A saline solution has mean concentration 0.900 M and standard deviation 0.005 M. A technician measures 0.912 M. Is this unusual?

Solution

$$z = \frac{0.912 - 0.900}{0.005} = 2.4$$

A z-score of 2.4 is well above 2 standard deviations from the mean, so this concentration is unusual under the model and should be checked.

Application: Comparing Different Scales

Question

A student scores 1250 on the SAT where the mean is 1050 and the standard deviation is 100. The same student scores 28 on the ACT where the mean is 21 and the standard deviation is 5. Which score is more impressive relative to its own distribution?

Solution

$$z_{SAT} = \frac{1250 - 1050}{100} = 2.0$$

$$z_{ACT} = \frac{28 - 21}{5} = 1.4$$

The SAT score has the larger z-score, so it is stronger relative to its reference distribution.

Application: Setting a Review Threshold for Claims

Question

Minor auto accident claims are modelled as Normal with mean 1500 dollars and standard deviation 200 dollars. What claim amount marks roughly the most expensive 2.5%?

Explanation and Solution

The upper 2.5% begins at about $z = 2$ under the 68–95–99.7 rule.

$$2 = \frac{x - 1500}{200}$$

$$x = 1900$$

Claims above about 1900 dollars fall in the upper 2.5% region of this approximate model.

Application: Laser Manufacturing Tolerances

Question

Laser wavelengths are modelled as Normal with mean 650 nm and standard deviation 2 nm. A sensor works safely for wavelengths from 644 nm to 656 nm. What percentage of lasers fall in this safe range?

Solution

The endpoints correspond to

$$z = \frac{644 - 650}{2} = -3$$

$$z = \frac{656 - 650}{2} = 3$$

The interval from -3σ to $+3\sigma$ contains about 99.7% of the distribution, so about 99.7% of lasers fall in the safe range.

Application: Setting a Warranty Threshold

Question

Refrigerator lifespans are modelled as Normal with mean 5.0 years and standard deviation 1.2 years. What warranty length leaves about 16% of refrigerators failing during the warranty period?

Explanation

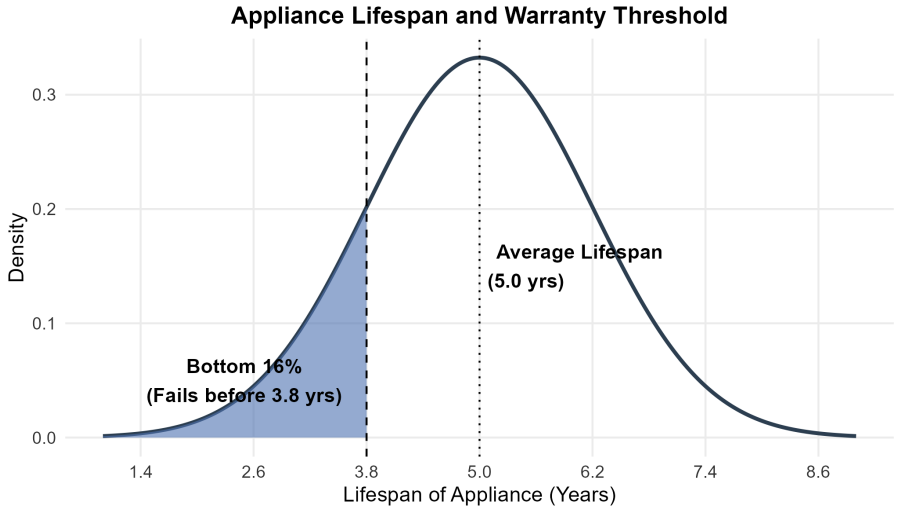
A left-tail area of 16% is approximately one standard deviation below the mean under the 68–95–99.7 rule.

The warranty boundary is

$$5.0 - 1.2 = 3.8 \text{ years}$$

A warranty of about 3.8 years leaves roughly 16% of refrigerators failing before that boundary under the model.

Application: Warranty Threshold Solution



Break Time



Cumulative Probabilities

The cumulative probability for a value x is the area at or below x.

$$P(X \leq x)$$

For continuous distributions, there is no difference between $P(X < x)$ and $P(X \leq x)$ because a single point has area 0.

Areas Under a Normal Curve as Probabilities

For a continuous probability model, area under the density curve represents probability.

If X follows a Normal distribution, then the probability that X is below a value x is the area to the left of x under the curve.

R calculates these probabilities with the function `pnorm()`.

```
pnorm(q, mean, sd)
```

Example 11.4: Bone Mineral Density and T-Scores

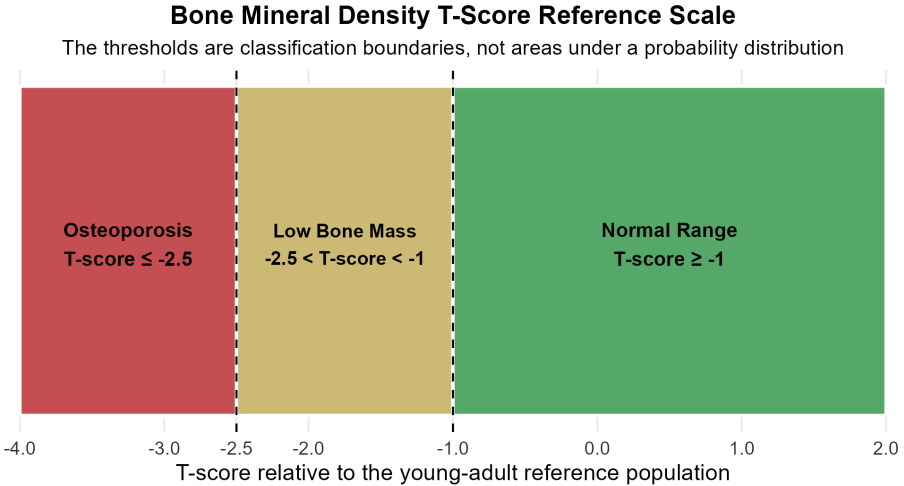
T-scores are standardized comparisons used in bone-density assessment.

Clinical interpretation often uses the following cutoffs.

- Normal: T-score at least -1.0
- Osteopenia: T-score between -2.5 and -1.0
- Osteoporosis: T-score at most -2.5

These cutoffs are reference thresholds. They do not imply that every patient population has a perfect standard Normal distribution.

Example 11.4: Bone Mineral Density and T-Scores



Example 11.4: Osteoporosis as a Left-Tail Probability

Question

Suppose women's T-scores in their 70s are approximated by a Normal model with mean -2 and standard deviation 1. What is the probability that a randomly selected woman has a T-score below -2.5?

Explanation

This is a left-tail probability. The boundary value is known, so `pnorm()` is appropriate.

The same left-tail probability is calculated in R with `pnorm()`.

```
# Left-tail probability
p_osteoporosis <- pnorm(q = -2.5, mean = -2, sd = 1)
print(p_osteoporosis)

>[1] 0.3085375
```

Example 11.4: Osteoporosis as a Left-Tail Probability

The probability statement is

$$P(X < -2.5) = 0.3085$$

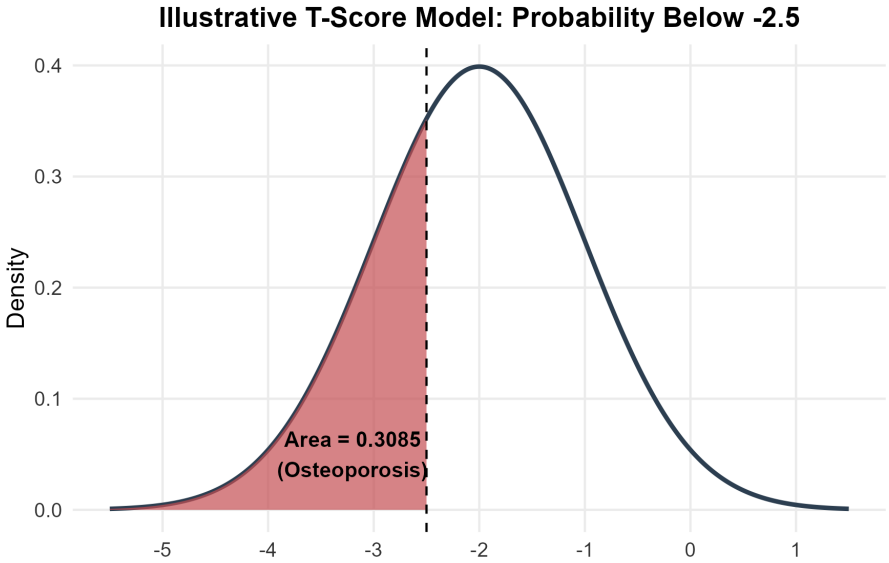
Under this illustrative model, about 30.85% of the distribution lies below the osteoporosis threshold.

The complement gives the proportion above the threshold

$$1 - 0.3085 = 0.6915$$

About 69.15% of the distribution lies above the threshold.

Example 11.4: Osteoporosis as a Left-Tail Probability



Example 11.4: Osteopenia as an Interval Probability

Question

Under the same T-score model, what proportion of women have osteopenia, defined as a T-score between -2.5 and -1?

Explanation

This is the area between two values. Since `pnorm()` gives cumulative left-tail probabilities, the lower cumulative area must be subtracted from the upper cumulative area.

```
# Area between -2.5 and -1
p_osteopenia <-
 pnorm(q = -1, mean = -2, sd = 1) - pnorm(q = -2.5, mean = -2, sd = 1)
print(p_osteopenia)
```

```
> [1] 0.5328072
```

The subtraction follows the same reasoning as the probability calculation: upper cumulative area minus lower cumulative area.

Example 11.4: Osteopenia as an Interval Probability

The required probability is

$$P(-2.5 < X < -1) = P(X < -1) - P(X < -2.5) = 0.8413 - 0.3085 = 0.5328$$

Using the model

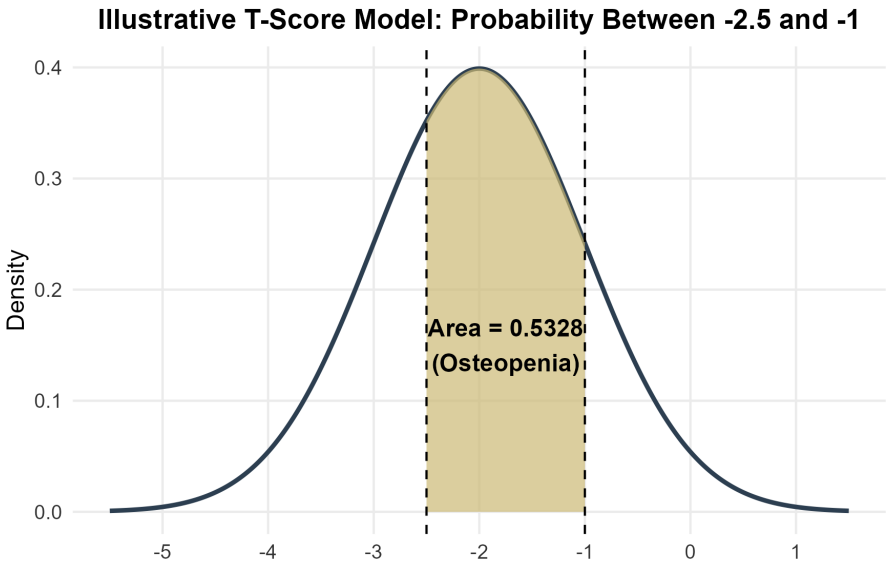
$$0.8413 - 0.3085 = 0.5328$$

Under this illustrative model, about 53.28% fall in the osteopenia interval. Note that

```
# Middle-tail probability
p <- pnorm(q = -1, mean = -2, sd = 1)
print(p)

>[1] 0.8413447
```

Osteopenia Interval Visualized



From Values to Probabilities and Back Again

Two R functions play different roles.

- `pnorm(q, mean, sd)` returns the probability of a value at or below q.
- `qnorm(p, mean, sd)` returns the value with probability p to its left.

What is a percentile?

The k**th percentile** is a value at or below which approximately k% of observations fall. For example, the 90th percentile of patient heights is the height at or below which 90% of patients fall. The remaining 10% are taller.

For a continuous Normal model, the 90th percentile is the boundary q satisfying

$$P(X \leq q) = 0.90$$

Example 11.6: The Third Quartile of Hatching Weights

Question

Commercial chicken hatching weights are modelled as Normal with mean 45 g and standard deviation 4 g. What is the third quartile of the distribution? That is, find the value satisfying

$$P(W \leq Q_3) = 0.75$$

Explanation

The third quartile is the 75th percentile, so the desired value has 75% of the distribution below it. This is a `qnorm()` question.

Example 11.6: The Third Quartile of Hatching Weights

The R code is

```
# The 75th percentile of the
# hatching-weight model
q3_weight <- qnorm(p = 0.75, mean = 45, sd = 4)
print(q3_weight)
```

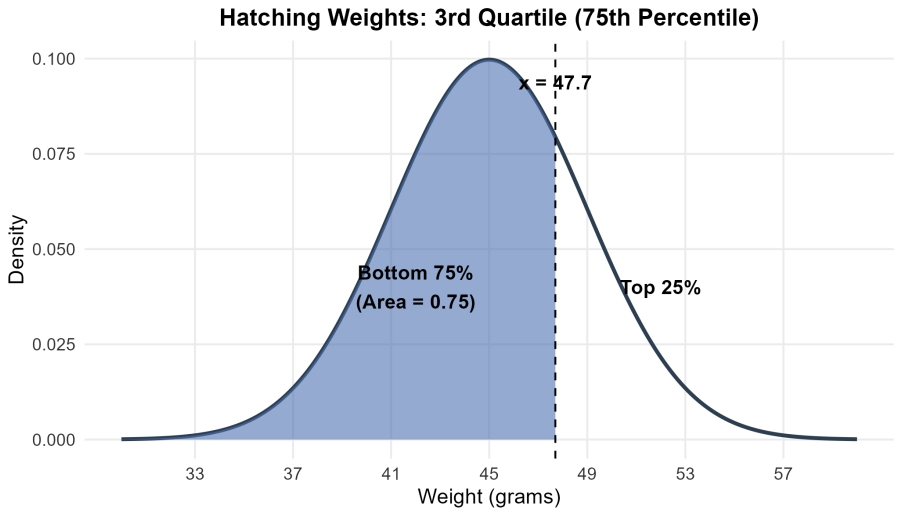
```
> [1] 47.69796
```

The result is approximately

$$47.7 \text{ g}$$

Under the model, about 75% of hatching weights are below 47.7 g and about 25% are above it.

Example 11.6: The Third Quartile of Hatching Weights



Example 11.7: The Longest 10% of Pregnancy Lengths

Question

Pregnancy lengths are modelled as Normal with mean 266 days and standard deviation 16 days. What pregnancy length marks the beginning of the longest 10%? That is, find q satisfying

$$P(D > q) = 0.10 \qquad \text{or equivalently} \qquad P(D \leq q) = 0.90$$

Explanation

The longest 10% means the upper 10%. Since `qnorm()` uses cumulative area to the left, the required boundary is the 90th percentile.

Example 11.7: The Longest 10% of Pregnancy Lengths

The R code is

```
# The 90th percentile leaves
# 10% above the boundary
pregnancy_90 <- qnorm(p = 0.90, mean = 266, sd = 16)
print(pregnancy_90)
```

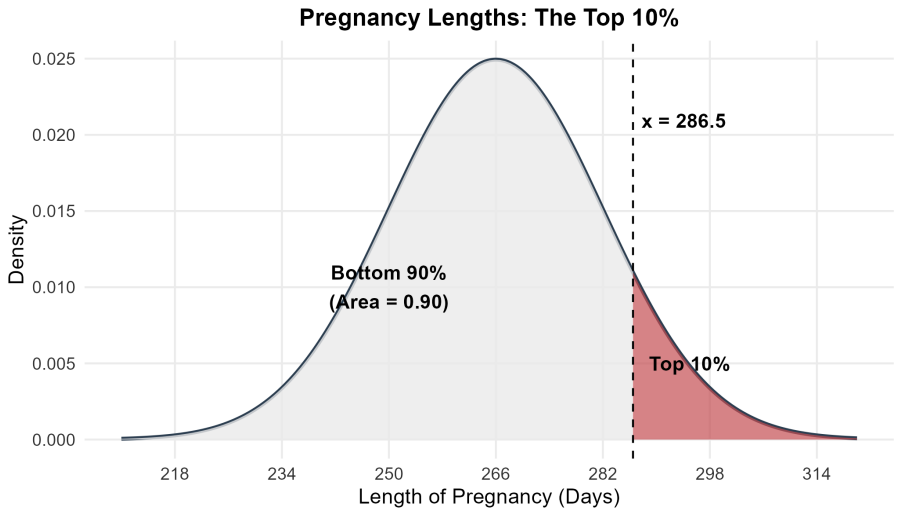
```
> [1] 286.5048
```

The result is approximately

286.5 days

If pregnancy length is reported to the nearest whole day, values of about 287 days or longer fall in the upper 10% region.

Example 11.7: The Longest 10% of Pregnancy Lengths



The Standard Normal Table

The Standard Normal table reports cumulative probabilities for

$$Z \sim N(0, 1)$$

Each table entry gives the area to the left of a specified z-score.
This is the same type of probability returned by `pnorm()` after a value has been standardized.

Reading the Z-Table

Question

What proportion of the standard Normal distribution lies below $z = 1.47$?

Explanation

Use row 1.4 and column 0.07. Their intersection gives the cumulative left-tail probability.

Solution

The table entry is 0.9292 so

$$P(Z < 1.47) = N(1.47) = 0.9292$$

About 92.92% of the distribution lies below 1.47.

Using a Standard Normal Table

Suppose $X \sim N(\mu, \sigma)$. Our table gives the cumulative **left-tail probability**, N(z) $= P(Z \leq z)$.

$$P(Z \leq z) = N(z)$$
$$P(Z > z) = 1 - N(z)$$
$$P(z_1 < Z < z_2) = N(z_2) - N(z_1) \qquad (z_1 < z_2)$$

1. **Identify the region.** Write the desired probability, such as $P(X > a)$ or $P(a < X < b)$, and sketch the corresponding area.
2. **Standardize each boundary.** Convert values from their original units to z-scores

$$z = \frac{x - \mu}{\sigma}$$

3. **Read the table and combine areas.** For $z = 1.23$, use row 1.2 and column 0.03. The entry gives $P(Z \leq 1.23) = 0.8907$.

If your table lists only positive z-scores, use symmetry $N(-z) = 1 - N(z)$.

Example 11.9: Heights of Patients with the Z-Table

Question

Patient's heights are modelled as Normal with mean 64.5 inches and standard deviation 2.5 inches. What is the probability that a randomly selected woman measures between 60 and 68 inches?

Explanation

Both boundaries must be standardized. The interval probability is then found by subtracting the cumulative area left of the lower z-score from the cumulative area left of the upper z-score.

Example 11.9: Heights of Patients with the Z-Table

The standardized boundaries are

$$z_1 = \frac{60 - 64.5}{2.5} = -1.8$$

$$z_2 = \frac{68 - 64.5}{2.5} = 1.4$$

Then

$$\begin{aligned} P(60 \leq X \leq 68) &= P(-1.8 \leq Z \leq 1.4) \\ &= N(1.4) - N(-1.8) \\ &= N(1.4) - \big(1 - N(1.8)\big) \\ &= 0.9192 - (1 - 0.9641) \\ &= 0.9192 - 0.0359 \\ &= \boxed{0.8833} \end{aligned}$$

Example 11.9: Interpretation and R Check

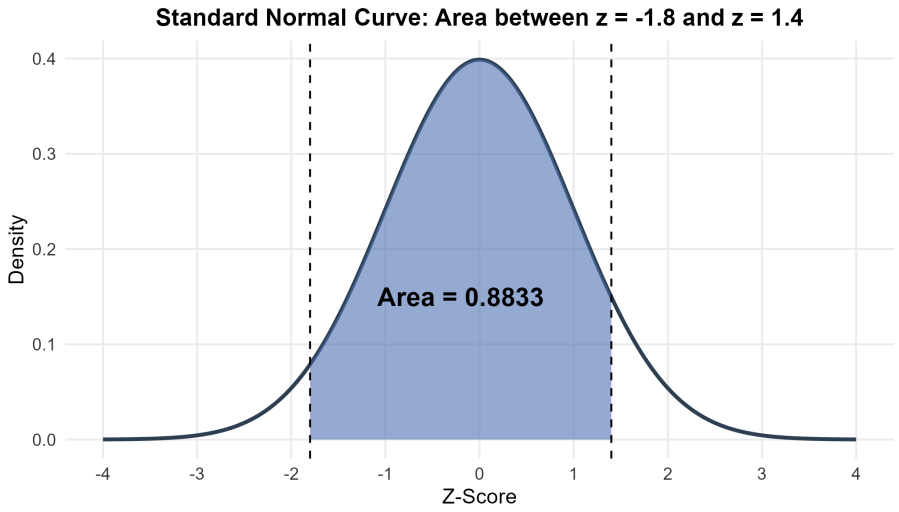
Under the Normal model, about 88.33% of young women measure between 60 and 68 inches.

The same probability can be checked in R.

```
# Probability between 60 and 68 inches
p_height_interval <-
 pnorm(68, mean = 64.5, sd = 2.5) - pnorm(60, mean = 64.5, sd = 2.5)
print(p_height_interval)
```

```
> [1] 0.883313
```

Interval Probability Visualized



Finding Percentiles with the Z-Table

The kth percentile is a boundary x satisfying

$$P(X \leq x) = \frac{k}{100}$$

We know the area to the left and want the corresponding value.

1. **Convert the percentile to a probability.** For the 90th percentile, the desired left-tail area is 0.90.
2. **Find the closest area inside the table.** Look for 0.9000 among the probability entries. The closest entry is 0.8997.
3. **Read the corresponding z-score.**That entry is in row 1.2 and column 0.08, giving $z = 1.28$. Thus, $P(Z \leq 1.28) \approx 0.90$.
4. **Convert back to the original units.** Use the distribution's mean and standard deviation

$$\boxed{x = \mu + z\sigma}$$

Example 11.10: The Longest Pregnancies Using the Z-Table

Let D be pregnancy length, in days:

$$D \sim N(266, 16)$$

Question

What boundary q marks the longest 10% of pregnancies? That is, find q satisfying

$$P(D > q) = 0.10 \qquad \text{or equivalently} \qquad P(D \leq q) = 0.90$$

Explanation

We need the z-score satisfying $P(Z \leq z) = 0.90$. The closest cumulative probability in the table is 0.8997, which corresponds to $z = 1.28$.

Example 11.10: The Longest Pregnancies Using the Z-Table

The boundary satisfies

$$P(D > q) = P\left(Z > \frac{q - 266}{16}\right) = 0.10$$

Using the table approximation $z = 1.28$

$$\frac{q - 266}{16} \approx 1.28$$

Solving for q gives

$$q \approx 266 + (1.28)(16) = \boxed{286.48 \text{ days}}$$

Interpretation

Under this model

$$P(D > 286.48) \approx 0.10$$

Approximately 10% of pregnancies last longer than 286.5 days.

Example 11.10: The Longest Pregnancies Using the Z-Table

We use `qnorm()`

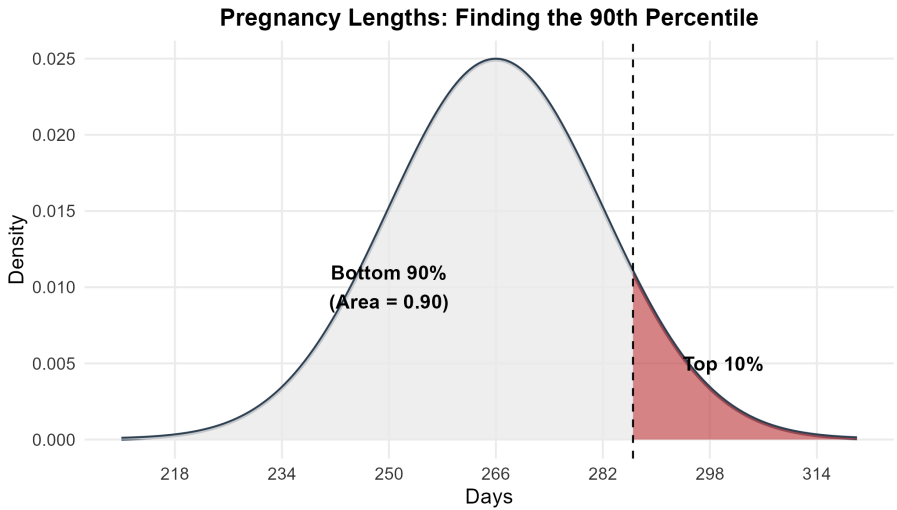
```
q <- qnorm(0.90, mean = 266, sd = 16)
q
```

```
> [1] 286.5048
```

Comparison with the Z-Table

R gives about 286.50 days. The table gives 286.48 days because it rounds the z-score to 1.28.

Example 11.10: The Longest Pregnancies Using the Z-Table



R Functions Used in Chapter 11

R Function	Purpose in This Chapter
`pnorm()`	Finds a cumulative Normal probability. It answers a value-to-probability question such as "What proportion lies below this value?"
`qnorm()`	Finds a Normal percentile or boundary. It answers a probability-to-value question such as "What value has 90% below it?"
`print()`	Displays a value stored in an R object so the numerical result can be inspected.

The Main Distinction

`pnorm()` goes from a **value to an area**. `qnorm()` goes from an **area to a value**.

R Code Patterns to Remember

A left-tail probability has the form

```
pnorm(q = boundary, mean = mu, sd = sigma)
```

An interval probability has the form

```
pnorm(upper, mean = mu, sd = sigma) - pnorm(lower, mean = mu, sd = sigma)
```

A percentile or boundary has the form

```
qnorm(p = cumulative_area, mean = mu, sd = sigma)
```

The probability reasoning should be established before the R function is selected.